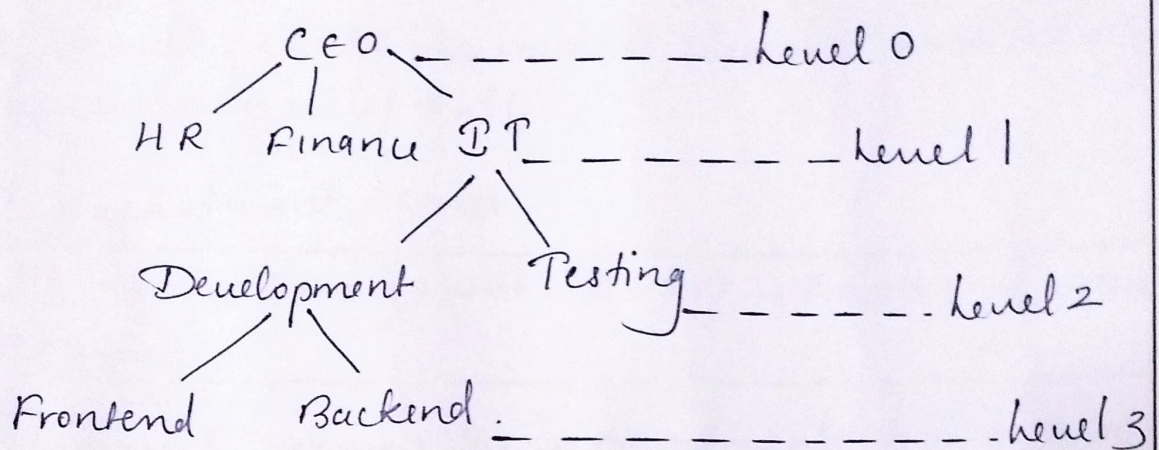


ANALYSIS & JUSTIFICATION.

Tree Structure Analysis:

The organisational tree is represented using a general tree because each department can have multiple child departments.



The hierarchy contains 4 levels and the height of the tree is 3 (or 4 levels).

Traversal Behaviour:

Level order traversing is used to display the organisational hierarchy. Level order traversal visits the nodes level by level from the root. The traversal uses Queue. Whenever a department is visited:

- Department get removed from queue from front.
- Get visited.
- Then puts its children at the back of queue, then visit
- Repeats the process.

Level order traversal output:

CEO → HR → Finance → IT → Development → Testing → Frontend
→ Backend.

Search Comparisons:

The departments are sorted in alphabetical order for suitable searching.

Linear search checks the element one by one from the beginning until the required department is found.

Binary search checks the middle element and reduces the search range by half.

Sorted department names:

Backend $\rightarrow$ CEO $\rightarrow$ Development $\rightarrow$ Finance $\rightarrow$ Frontend $\rightarrow$ HR $\rightarrow$ IT $\rightarrow$ Testing.

Binary search requires the department names to be sorted. The same sorted array is therefore used for linear search.

Search	Linear Search comparisons	Binary Search comparisons
Development	Backend $\rightarrow$ CEO $\rightarrow$ Development $= 3$	Finance $\rightarrow$ CEO $\rightarrow$ Development $= 3$.
HR	Backend $\rightarrow$ CEO $\rightarrow$ Development $\rightarrow$ Finance $\rightarrow$ Front end $\rightarrow$ HR $= 6$	Finance $\rightarrow$ HR. $= 2$.
IT	Backend $\rightarrow$ CEO $\rightarrow$ Development $\rightarrow$ Finance $\rightarrow$ Frontend $\rightarrow$ HR $\rightarrow$ IT $= 7$	Finance $\rightarrow$ Frontend $\rightarrow$ HR $\rightarrow$ IT $= 4$.

Time Complexity:

- Tree Construction - $O(n)$:

The tree is created by connecting each node with its parents. So when the number of nodes increases, work required increases. Therefore time complexity is $O(n)$.

- Level Order Traversal - $O(n)$

Level order traversal traverses every node in tree exactly once. Therefore for n number of nodes traversal takes place n times so $O(n)$.

- Linear Search - $O(n)$.

This searching method checks the department names one by one from the beginning of the array, so all n nodes have to be checked therefore $O(n)$.

- Binary Search - $O(\log n)$.

In binary search, it checks the middle element and eliminates approximately half of the remaining elements after each comparison. Therefore number of elements to be searched decreases rapidly, giving worst case complexity $O(\log n)$ and it works on sorted arrays.

Operations	Time Complexity
• Tree Traversal	$O(n)$
• Level-order Traversal	$O(n)$
• Linear Search	$O(n)$
• Binary Search	$O(\log n)$

Final Justification & Conclusion:

The general tree representation is suitable for organisational reporting because it represents the parent-child relationship between CEO and the departments clearly. The level-order traversing displays the organisation level by level, making the hierarchy easy to understand.

The department names are sorted away, which is suitable for department searching. Linear search can locate the department by checking name one by one while Binary search can locate a department by repeatedly reducing the search range by half. The comparison result shows that Binary search can require fewer comparisons than linear search. The time complexities also support the use of these representations.

Together, these representations satisfy the requirement of organisational reporting and department searching.